

NEUR 411 Neuroethology

Spring 2027

Instructor

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Course Times & Locations

Component	Days & Times	Location
Lecture, Section 1	Mon / Wed / Fri, 9:00 to 10:50 AM	NEU 500
Lecture, Section 2	Mon / Wed / Fri, 5:30 to 7:20 PM	NEU 500
Laboratory	Lab Sections: Tue 11am – 1pm Tue 4pm – 6pm Wed 12pm – 2pm Thu 3pm – 5pm Fri 12pm – 2pm	 NEU LAB5A NEU LAB5B NEU LAB5A NEU LAB5B NEU LAB5A
Exams (4 total)	Mondays at 12:00 PM (dates in Course Calendar)	NEU 100A or NEU 100B

Course Description

Neuroethology sits at the intersection of neuroscience, ethology, and evolutionary biology. It asks how nervous systems generate the behaviors that allow animals to find food, avoid predators, navigate across landscapes, and communicate with conspecifics, and how those neural solutions have been shaped by natural selection. NEUR 411 examines this question through a series of case studies, each centered on a single animal system: insect vision and hearing, the escape circuits of crayfish and fish, bat and owl sound localization, electroreception in sharks and weakly electric fish, magnetic and chemical navigation, songbird vocal learning, and primate communication. Within each case, we consider the behavior the animal must perform, the neural circuitry that performs it, and the experiments that established the mechanism.

The course is organized around Tinbergen's four questions (causation, development, function, and evolution), which provide a framework for distinguishing how a behavior works from how it develops, why it persists, and how it arose. Students will learn to evaluate the experimental evidence behind these claims, to recognize the limits of that evidence, and to compare how different taxa solve the same ecological problem. The course draws on more than 1,000 articles reviewed in designing the curriculum.

As the field evolves rapidly, this course is structured to cover as much current research as possible within a demanding schedule. Students should expect a substantial and sustained workload. By the end of the semester, students should be able to articulate, with supporting evidence, the neural capabilities that allow animals to detect prey, localize sounds, navigate long distances, and communicate with one another, and to assess how well current models explain those capabilities.

Required Field Experience: Spring Recess

The Spring Recess Field Experience is a required, four-day neuroethology field trip from Thursday, March 18 through Sunday, March 21, 2027. The trip is educational. You will evaluate animal welfare and behavior scientifically, using the methods and concepts from lecture. It is not recreational tourism. Your job is to observe, critique, and understand what animals' environments and behaviors show about sensory processing, learning, stress, and enrichment. The host sites know we are coming and are prepared for us. There is no cost to students. All expenses are paid by the program. The trip is mandatory, and attendance on all four days is required.

- **Thursday, March 18:** Beckman Center for Conservation Research & San Diego Zoo Safari Park (Escondido, CA).
- Special topic: *Animal Behavior in Managed Environments I: Zoos (sensory ecology, neural mechanisms of stress, social behavior, enrichment, and neuroethical considerations of captivity).*

- **Friday, March 19:** Hubbs-SeaWorld Research Institute *&* SeaWorld Rescue & Rehabilitation Center (San Diego, CA). **We will carry out a neuroethical critique and behavioral audit.**
- *Special topic: Animal Behavior in Managed Environments II: Aquariums & Marine Theme Parks (underwater sensory environments, learning and reward systems, social communication, and neuroethical considerations of training and performance).*

- **Saturday, March 20:** Aquarium of the Pacific (Long Beach).
- *Special topic: Animal Behavior in Managed Environments III: Aquarium Sensory Worlds (visual, chemical, and mechanosensory processing, sensory adaptations, and how exhibit design shapes natural behavior)*

- **Sunday, March 21:** Marine Mammal Care Center for a few hours. Then head back to USC around 1-2pm
- *Special topic: Animal Behavior in Managed Environments IV: Rescue, Rehabilitation & Release (neurobiology of stress, behavioral/injury recovery, human habituation, and assessment of readiness for release)*

Overnight stays. We will stay overnight at the trip locations on each night of the trip, including the Aquarium of the Pacific. Bring a sleeping bag and whatever you need to sleep well, such as a sleeping pad, memory foam topper, pillow, and warm layers. The department will provide a limited number of sleeping bags from prior trips, but bring your own if you can.

Meals and schedule. Meals are provided by the program: breakfast at 9:00 AM, lunch at 1:00 PM, and dinner at 6:00 PM. For most of the day, you will have time to explore the sites and to learn directly from field biologists. After dinner, you will have free time.

Exams and evaluation. Field trip material will not be covered on exams. It will be evaluated through the Captivity and Conservation Reflection.

Absences. The trip is mandatory. An absence is excused only with documentation approved in advance. If you cannot attend or anticipate a conflict, contact me as soon as possible. The trip is costly and is planned in advance, so do not wait.

These four days appear in the Course Calendar as special topic sessions. We will meet as a class during the trip for informal discussions, animal observations, and group activities, with time to explore and enjoy the visits.

Participation is required on all four days (attendance will be taken and incorporated into your lab participation grade). No assignments are due during the trip

Prerequisites (Enforced)

[NEUROBIOLOGY PREREQUISITE & STATS COURSE/INTRODUCTORY STATISTICS]. Both are required. I will verify them at enrollment, and students without both will be dropped. You must be able to read a figure with error bars, interpret a significance test or regression output, and reason about cellular and systems neurophysiology from primary papers without being walked through each step. If you are hoping to catch up in the first three weeks, withdraw now.

Course Materials

There is no required textbook. Lecture slides are the primary source of instruction, and exams are based on lecture content. Recommended textbook: Zupanc, G. K. H. *Behavioral Neurobiology: An Integrative Approach* (3rd ed.). Oxford University Press, ISBN 978-0-19-873872-5. Alternative: Carew, T. J. (2000). *Behavioral Neurobiology: The Cellular Organization of Natural Behavior.* Sinauer.

Readings are listed in the Readings Calendar after the Course Calendar. They are recommended, not required. I will teach the material in lecture, and the readings are provided so that you can go deeper and succeed on exams.

Required equipment: a laptop capable of running Python or MATLAB for the simulation project, and a reliable connection to the course site. Software is free to install; instructions will be posted in Week 1.

Course Objectives

- Explain how sensory systems extract ecologically relevant information from the environment, and how that information is converted into motor output.
- Describe neural circuit mechanisms (identified neurons, central pattern generators, sensory maps, and neuromodulation) that generate specific animal behaviors.
- Evaluate experimental evidence from behavioral, electrophysiological, anatomical, and lesion studies, including its limitations.
- Compare solutions that different taxa use for the same problem, including convergent evolution.
- Design and communicate a quantitative analysis or model of a natural behavior using lab data or published results.

Teaching Assistants

This course has no co-instructor. Questions about lab logistics, lab reports, and grading should go to your TA first. Email the instructor only for issues that the TA cannot resolve.

Role	Name	Email	Office Hours / Location
Lab TA (Section M, TH)	Nicholas Xe	nicxe@usc.edu	Tue 1:00–3:00 PM, Thu 10:00 AM–12:00 PM, HNB107
Lab TA (Section 2)	Xavier Guden Lopez	Xavlop@usc.edu	Wed 2:30–4:30 PM, Fri 2:30–4:30 PM, HNB107
Exam Proctor	Dr. Luis Lemus Dr. Yahaira S Sanchez	lemus@usc.edu saraiy@usc.edu	Mondays, NEU 100A, 12:00 PM Mondays, NEU 100B, 12:00 PM

Grading

This course has two components, lecture and laboratory. Each is scored out of 100 percent. Your final grade is 50 percent lecture and 50 percent laboratory. **You must earn at least 60 percent in BOTH components to pass the course.** A strong lecture score does not rescue a failing lab score, and the reverse is also true.

Lecture Component (100%)

Assessment	Weight of Lecture Grade	Points
Exam 1 (Lectures 1 to 10)	20%	120
Exam 2 (Lectures 11 to 20)	20%	120
Exam 3 (Lectures 21 to 30)	20%	120
Exam 4 (Lectures 31 to 40)	20%	120
Lecture Project (simulation or APA paper)	20%	120
Total	100%	600

Laboratory Component (100%)

Assessment	Weight of Lab Grade
Lab Reports (8 counted of 10, 7.5% each)	60%
Lab Attendance and Participation	10%
Computational Assignments (3 at 5% each)	15%
Captivity and Conservation Reflection (5 pages)	15%
Total	100%

Dr. J’s Grading Philosophy and Curve

Neuroethology requires you to connect neural mechanisms with animal behavior, interpret experimental evidence, and compare findings across species. This is a demanding upper-division course. Memorizing terminology alone is not sufficient; you must be able to explain how a circuit works, evaluate the evidence supporting a proposed mechanism, and apply your understanding to unfamiliar experimental results.

Exams and laboratory assignments reflect these expectations. Some assessments will be difficult, and raw percentages may be lower than you are accustomed to. Grades reflect demonstrated understanding, scientific accuracy, and the quality of your reasoning. Consistent effort and participation are important, but they do not substitute for meeting the course’s academic standards.

If the final class average, calculated using the weighted lecture and laboratory grades before curving, is below 70%, a curve will be applied at the end of the course. If the class average is 70% or higher, no curve will be applied. The size of any adjustment will depend on overall class performance. A curve accounts for assessment difficulty while maintaining expectations for mastery; it does not guarantee a particular letter grade or a passing grade. Use early assessment results to identify gaps in your understanding, review feedback, and seek help promptly. Do not rely on a possible curve to compensate for missing work or consistently weak performance. You must still earn at least 60% in both the lecture and laboratory components to pass the course.

Grading Scale After Curving (changes year to year)

The percentage ranges below are approximate estimates, not fixed or guaranteed cutoffs. The grading scale may change from year to year, and final cutoffs for this semester may differ based on overall class performance and any curve applied under the policy above. Use this table as a general guide when estimating your grade.

Letter Grade	Approximate Percentage Range
A	90-100%
A-	85-89.9%
B+	80-84.9%
B	75-79.9%
B-	70-74.9%
C+	65-69.9%
C	60-64.9%
C-	55-59.9%

Letter Grade	Approximate Percentage Range
D	45-54.9%
F	Below 45%

Final grades are not rounded up. Letter grades will be assigned strictly according to the finalized cutoffs, which will be announced before final grades are assigned. Regardless of any adjustment to the scale, you must earn at least 60% in both the lecture and laboratory components to pass the course.

Exams

There are four exams, each worth 120 points, and each covers exactly ten lectures. Exam 1 covers Lectures 1 to 10, Exam 2 covers Lectures 11 to 20, Exam 3 covers Lectures 21 to 30, and Exam 4 covers Lectures 31 to 40. Exams are not cumulative, but the concepts build on one another, so later exams will assume you still know earlier material. Exams are given on Mondays at 12:00 PM in NEU 100A or NEU 100B, and they start on time. Exam 4 is given during the final examination period. On the days of Exams 1, 2, and 3, the lecture is a special topic. Each special topic may appear as 2 to 4 questions on the next exam (Special Topic 1 on Exam 2, Special Topic 2 on Exam 3, and Special Topic 3 on Exam 4), but special topics are not a major emphasis.

- **Number of questions and versions:** Expect 80 questions on each exam. Each exam comes in three versions (V1, V2, and V3). The three versions contain the same questions, with the order of questions and answer choices mixed within each lecture section.
- **Question order:** Exam questions are ordered by lecture. Questions on Lecture 1 come first, followed by questions on Lecture 2, and so on through the last lecture covered by that exam.
- **Room and version assignment:** Before each exam, a randomized algorithm assigns every student to Room A (NEU 100A) or Room B (NEU 100B) and to one of three versions (V1, V2, or V3). Assignments are posted on the course site before the exam. Assignments are random, and requests to change them will not be accepted.
- **Exam format: In Person, Paper Exams** consisting primarily of written free-response questions and fill-in-the-blank questions, with a small number of multiple-choice questions. Written questions may require short answers, longer explanations, interpretation of figures and data, experimental design critiques, and cross-species comparisons.
- **Expectations for answers:** You must draw directly from lecture content, explain specific experimental findings, and provide relevant examples of animals, behaviors, and neural mechanisms. When asked to support an explanation, identify the evidence and explain how it supports your answer. Memorizing terminology or repeating slide text without demonstrating understanding is not sufficient.
- **Missed exams:** Make-up exams are not given. If you miss one exam, a zero will initially be recorded. If the absence is approved for a valid, documented medical or official reason, that zero will be replaced at the end of the semester by the average of your other three exam scores. This replacement uses your other exam scores, not your final course grade. An undocumented or unapproved absence will remain a zero.
- Bring a photo ID and a pencil (YOU WILL BE VERIFIED). Phones, smartwatches, and notes are not allowed
- **Exams are closed-note** unless the class negotiates an alternative at the beginning of the semester (and/or before Exam I). The class may negotiate permission to bring **one 8.5 × 11-inch handwritten reference sheet (double sided, OK). In exchange, the class agrees to replace the multiple-choice**

questions with additional fill in the blank questions (any change needs to be negotiated with the class as a whole). Reference sheets must be handwritten; typed or printed material is not permitted.

New lecture content is introduced on exam days. These lectures are shorter, but attendance is required, and the material will be covered on the next exam.

Lecture Project (20% of Lecture Grade)

Each student completes one project based on an animal system covered in class. Choose ONE option. Your choice is due with your proposal.

Option A: Simulation Project

Build a computational simulation of a neural circuit, sensory system, or behavior from one animal covered in lecture (for example, the cricket phonotaxis circuit, the Mauthner startle circuit, a CPG, or the owl ITD map). Models must be implemented in Python, MATLAB, or an equivalent tool. Deliverables: the working code (commented and runnable), a 5-page written report with the model's assumptions, parameter justification, validation against published data, and a discussion of where it fails, and a 5-minute recorded walkthrough. A model that cannot be compared with published data will not earn more than 70 percent of the project points.

Option B: 10-Page APA Paper

Write a 10-page paper in APA 7th edition format, excluding title page, abstract, and references, on one animal covered in class. The paper must synthesize at least 8 primary research articles, explain the neural mechanism and the behavior it produces, and critically evaluate the evidence. Summaries of individual papers will receive low marks. The paper must argue something.

Project Milestones

Milestone	Due
Proposal: animal, option chosen, one research question, and key references (1 page)	Mon Mar 29, 11:59 PM
Progress check: outline or working model with first results	Wed Apr 14, 11:59 PM
Final project submission	Fri May 7, 11:59 PM

Laboratory Component

The lab is a required, graded component and is not optional. Labs use recorded and published data, preserved specimens, and, where approved, live animal observation carried out under an institutional animal care protocol. You will complete animal welfare and lab safety training before Lab 1. Mishandling of animals, data, or equipment results in removal from the lab and a failing lab grade.

- **Lab Reports (10 labs, lowest 2 dropped, 7.5% each for the 8 counted).** Each report follows a standard format: Introduction, Methods, Results (with figures and statistics), Discussion, and References. Reports must be to 6-10 pages. Every quantitative claim requires a statistical test and an interpretation.
- **Lab Attendance and Participation (10%).** Attendance is required at every lab. Participation is earned by submitting your observations and data to Gradescope before you leave the lab. Submissions made after you leave will not be marked, and you will receive no participation credit for that lab. Excused absences require documentation submitted to the TA within 24 hours. Each unexcused absence reduces this grade by 15 points out of 100.
- **Computational Assignments (3 assignments, 5% each, 15% total).** Short Python or MATLAB problems covering data analysis and simulation that build the coding skills required for the simulation project. Due dates are in the schedule below.
- **Captivity and Conservation Reflection (15%).** A 5-page essay on your experience during the Spring Recess Field Experience and what you learned. Describe what you observed, evaluate the welfare and behavioral evidence scientifically, and explain what the visit taught you about how captive environments

shape behavior. The essay is due Sat Apr 24. It is the best opportunity in the course to show you understand the biology, and it is also the one most students skip. Do not skip it.

- **Research use of this essay:** This essay is part of research on young students participating in active research, studying learning accessibility and disparities. Use of your essay in that research is voluntary. Declining has no effect on your grade or on how your essay is scored. This statement applies only to the Captivity and Conservation Reflection.

Lab Schedule and Due Dates

Lab	Week	Topic	Due / Notes
Lab 1	Week 1	Ethograms and Tinbergen's four questions: quantitative observation of zebrafish behavior from recorded video	Report due Sun Jan 24
Lab 2	Week 2	Cockroach cercal system: directional wind tuning of giant interneurons from published recordings	Report due Sun Jan 31
Lab 3	Week 4	Stomatogastric pyloric rhythm: burst timing, phase constancy, and neuromodulation from published recordings	Report due Sun Feb 14
Lab 4	Week 5	Toad prey-catching: worm versus anti-worm stimulus tuning from recorded behavior	Report due Sun Feb 21
Lab 5	Week 6	Jumping spider visual tracking: saccades and pursuit from video	Report due Sun Feb 28
Lab 6	Week 8	Moth hearing and bat evasion: auditory receptor responses to ultrasound and evasive flight	Report due Sun Mar 14
Lab 7	Week 9	Weakly electric fish: electric organ discharge recordings and active electrolocation	Report due Mon Mar 29
Lab 8	Week 12	Rapid prey capture: star-nosed mole handling times and archerfish shot geometry from high-speed video	Report due Sun Apr 11
Lab 9	Week 13	Orientation and navigation: circular statistics on monarch butterfly and desert ant headings	Report due Sun Apr 18
Lab 10	Week 14	Songbird song structure: syllable segmentation and song stereotypy in zebra finch recordings	Report due Sun Apr 25
Make-up	Week 16	Make-up lab for students who missed a scheduled lab	-
Computational Assignment 1	Thu Feb 4	Python fundamentals, loading and plotting recordings, spike detection	Due 11:59 PM
Computational Assignment 2	Thu Mar 11	Signal analysis: power spectra and cross-correlation	Due 11:59 PM
Computational Assignment 3	Thu Apr 15	Simulation: a leaky integrate-and-fire neuron	Due 11:59 PM
Reflection Essay	Sat Apr 24	Captivity and Conservation Reflection (5 pages)	Due 11:59 PM

Lab section times and rooms are posted on the course site. The late work policy applies (see Late Work below).

Lecture Recordings and Study Materials

Lectures are recorded. Recordings are released **one week before the exam that covers them**. I may release them two weeks before an exam at my discretion, but early release is not guaranteed and should not be expected. You are responsible for attending lecture, taking thorough notes, and studying throughout the semester. **Do not wait for recordings to begin preparing for an exam.**

Recordings from previous offerings may also be posted as supplemental study materials. This course changes between semesters, and older recordings may omit current topics or include material that is no longer covered. **This semester's lectures determine what is examinable.** You are responsible for identifying those differences using the current lectures and slides.

Recordings do not replace attendance or excuse arriving unprepared. Lecture content is dense and requires consistent engagement. Choosing to rely on recordings does not reduce the material you are expected to know or change the standards by which your work will be graded.

Study Expectations

This course covers a lot of material on a fast schedule, and it will punish procrastination. Plan at least two hours of outside study for every hour of class. Review each lecture within 48 hours. Read the recommended papers for each lecture in the Readings Calendar. Work through the figures until you can explain the data without looking at the caption. Form study groups but make sure each of you can answer exam questions alone. Do not wait until the week of an exam to begin studying, and do not start the project the week it is due.

Lecture & Lab Attendance

Attendance at all lectures and labs is expected. Lectures introduce material that is not on the slides, and that material will appear on exams. There is no grade for lecture attendance, but missing lectures will affect your exam performance. Lab attendance is graded, and lab absences will not be excused without documentation.

Late Work

- **Lab reports, computational assignments, and the Captivity and Conservation Reflection:** 5% deduction per day late for the first 7 days. Work more than 1 week late is not accepted.
- **Lab participation:** observations and data must be submitted to Gradescope before you leave the lab. Submissions after you leave are not marked.
- **Lecture project:** 30% deduction for any submission after 11:59 PM on the due date. Projects submitted more than 48 hours late are not accepted.

Extensions are granted only for documented medical or emergency circumstances, and only if you contact me before the deadline. Asking for an extension the night before a deadline is not an emergency.

Academic Integrity and Use of AI Tools

Follow the university's standards of academic integrity. Cheating, plagiarism, unauthorized collaboration, and the use of unauthorized materials on exams will result in a zero on the work and a referral under university policy. Generative AI tools may not write any part of the lab reports, Captivity and Conservation Reflection, lecture project paper, or exam answers. You may use AI tools for coding assistance on the simulation project only if you disclose the use in your methods section and reference. ***Undisclosed use is a violation.***

Accommodations

Students with documented disabilities or other needs should contact the Office of Student Accessibility Services as early as possible. Once approved, provide me with your letter of accommodations so I can put it on file. Accommodations are honored only once they are approved and on file, and requests must be submitted at least 2 weeks prior to an exam.

Communication

Because this course involves complex, interdisciplinary concepts related to neural mechanisms and animal behavior, questions and clarification needs are expected. Students are strongly encouraged to use **Teaching Assistants as their first point of contact** for questions related to homework, labs, coding issues, and course logistics. TAs hold regular lab sessions and office hours and are well positioned to provide timely support. For questions that cannot be resolved with a TA, or for concerns related to unresolved grading disputes, accommodations, or broader course issues, students may contact the instructor at **jlopez11@usc.edu**.

I will make every effort to respond to emails and messages promptly; however, because I teach multiple courses and run an active research laboratory, response times may be delayed. Students should generally expect responses within **one business day**. Messages sent in the evening, overnight, or on weekends will typically be addressed the following business day. Use clear subject lines that include the course number and topic, such as “NEUR 411: Lab 3 report question.” Keep questions concise and include relevant context, such as the assignment

name or lab section. **Reach out early rather than waiting until deadlines or exam days.** Consult the syllabus and current lecture materials for exam coverage before emailing.

Absences and Communication

I understand that many students balance academic responsibilities with work, family obligations, and other commitments. Communication is key. If you anticipate an absence or experience a documented emergency, **please email me as soon as possible**. For lab absences, also notify your TA and submit documentation within 24 hours, as required by the laboratory attendance policy. Being proactive and communicative helps me support you throughout the semester.

Unexcused lab absences will affect your Lab Attendance and Participation grade under the laboratory attendance policy, regardless of access to lecture recordings. There is no separate lecture attendance grade, but you remain responsible for all missed lecture content. Recordings are provided as a supplement and do not replace in-person participation. Missed exams remain subject to the missed-exam policy.

Course Calendar

All dates are final unless announced in class and posted on the course site. Lectures meet Monday, Wednesday, and Friday, 1 hour and 50 minutes per meeting. Exams are Mondays at 12:00 PM in NEU 100A or NEU 100B. Lab meeting weeks and report due dates are listed in the Lab Schedule above.

Date (2027)	Session	Topic	Assessment / Notes
Week 1	Lab 1	Lab 1 meets: ethograms and Tinbergen's four questions: quantitative observation of zebrafish behavior from recorded video	*Must have approval by IACUC before entering lab*
Mon Jan 11	Lecture 1	Introduction to neuroethology: Tinbergen's four questions; proximate vs. ultimate causation	Will NOT review Syllabus (review on your own)
Wed Jan 13	Lecture 2	Fixed action patterns and sign stimuli: Lorenz and Tinbergen's stickleback and graylag goose work	
Fri Jan 15	Lecture 3	Cockroach escape: the cercal system and giant interneurons (Camhi)	
Week 2	Lab 2	Lab 2 meets: cockroach cercal system: directional wind tuning of giant interneurons from published recordings	
Mon Jan 18	No class	Martin Luther King Jr. Day	
Wed Jan 20	Lecture 4	Crayfish tail-flip: lateral and medial giant fiber circuits (Wine & Krasne)	
Fri Jan 22	Lecture 5	Fish startle: the Mauthner cell and the C-start response (Eaton & Faber)	
Sun Jan 24	Lab 1	Lab 1 report due	Report due
Mon Jan 25	Lecture 6	Central pattern generators I: leech swimming and crawling (Kristan)	
Wed Jan 27	Lecture 7	Central pattern generators II: the lobster/crab stomatogastric ganglion (Marder & Selverston)	
Fri Jan 29	Lecture 8	Locust flight: wind-sensitive interneurons and the flight CPG	
Sun Jan 31	Lab 2	Lab 2 report due	Report due
Week 4	Lab 3	Lab 3 meets: stomatogastric pyloric rhythm: burst timing, phase constancy, and neuromodulation from published recordings	
Mon Feb 1	Lecture 9	Amphibian prey capture I: toad visual prey-catching and tectal feature detectors (Ewert)	

Date (2027)	Session	Topic	Assessment / Notes
Wed Feb 3	Lecture 10	Amphibian prey capture II: pretectal predator-avoidance circuits	
Thu Feb 4	Comp. 1	Computational Assignment 1: python fundamentals, loading and plotting recordings, spike detection	Due 11:59 PM
Fri Feb 5	Lecture 11	Insect vision I: the fly optomotor response and the Reichardt elementary motion detector	
Week 5	Lab 4	Lab 4 meets: toad prey-catching: worm versus anti-worm stimulus tuning from recorded behavior	
Mon Feb 8	Special Topic 1	Jellyfish I: neuroanatomy of Cassiopea, distributed nerve nets, rhopalia, and pacemaker control of rhythmic bell contractions	2–4 questions on Exam 2
Mon Feb 8	**Exam 1**	**Covers Lectures 1 to 10**	**12:00 PM, NEU 100A or NEU 100B**
Wed Feb 10	Lecture 12	Insect vision II: jumping spider eyes and predatory tracking (Land)	
Fri Feb 12	Lecture 13	Mantis shrimp vision: hyperspectral and polarization sensing	
Sun Feb 14	Lab 3	Lab 3 report due	Report due
Week 6	Lab 5	Lab 5 meets: jumping spider visual tracking: saccades and pursuit from video	
Mon Feb 15	No class	Presidents' Day	
Wed Feb 17	Lecture 14	Insect hearing I: cricket phonotaxis and song recognition (Hoy)	
Fri Feb 19	Lecture 15	Insect hearing II: moth ears and bat-evasion behavior (Roeder)	
Sun Feb 21	Lab 4	Lab 4 report due	Report due
Mon Feb 22	Lecture 16	Bat echolocation I: CF-FM bats and Doppler-shift compensation (Suga)	
Wed Feb 24	Lecture 17	Bat echolocation II: FM bats, target-range computation, clutter rejection	
Fri Feb 26	Lecture 18	Owl sound localization I: ITD/ILD maps in the inferior colliculus (Konishi & Knudsen)	
Sun Feb 28	Lab 5	Lab 5 report due	Report due
Week 8	Lab 6	Lab 6 meets: moth hearing and bat evasion: auditory receptor responses to ultrasound and evasive flight	
Mon Mar 1	Lecture 19	Owl sound localization II: developmental plasticity of the auditory map	
Wed Mar 3	Lecture 20	Electroreception I: passive sensing in sharks and rays (ampullae of Lorenzini)	
Fri Mar 5	Lecture 21	Electroreception II: active electrolocation in weakly electric fish (Bullock)	
Week 9	Lab 7	Lab 7 meets: weakly electric fish: electric organ discharge recordings and active electrolocation	
Mon Mar 8	Special Topic 2	Axolotl: lateral-line sensing, electroreception, and sensory processing of courtship signals	2–4 questions on Exam 3
Mon Mar 8	**Exam 2**	**Covers Lectures 11 to 20 (and 2–4 questions on Special Topic 1)**	**12:00 PM, NEU 100A or NEU 100B**
Wed Mar 10	Lecture 22	Electroreception III: the jamming avoidance response (Heiligenberg)	
Thu Mar 11	Comp. 2	Computational Assignment 2: signal analysis: power spectra and cross-correlation	Due 11:59 PM
Fri Mar 12	Lecture 23	Magnetoreception: sea turtle and migratory bird magnetic compasses	
Sun Mar 14	Lab 6	Lab 6 report due	Report due
Mon Mar 15	No class	Spring Recess (no classes March 15–19)	

Date (2027)	Session	Topic	Assessment / Notes
Thu Mar 18	Special Topic Location	Animal Behavior in Managed Environments I: Zoos (sensory ecology, neural mechanisms of stress, social behavior, enrichment, and neuroethical considerations of captivity). *Beckman Center for Conservation Research & San Diego Zoo Safari Park*	Field Experience (required)
Fri Mar 19	Special Topic Location	Animal Behavior in Managed Environments II: Aquariums & Marine Theme Parks (underwater sensory environments, learning and reward systems, social communication, and neuroethical considerations of training and performance) *Hubbs Research Institute & SeaWorld Rescue & Rehabilitation Center*	Field Experience (required)
Sat Mar 20	Special Topic Location	Animal Behavior in Managed Environments III: Aquarium Sensory Worlds (visual, chemical, and mechanosensory processing, sensory adaptations, and how exhibit design shapes natural behavior) *Aquarium of the Pacific*	Field Experience (required)
Sun Mar 21	Special Topic Location	Animal Behavior in Managed Environments IV: Rescue, Rehabilitation & Release (neurobiology of stress and injury, behavioral recovery, human habituation, and assessment of readiness for release) *Marine Mammal Care Center*	Field Experience (required)
Mon Mar 22	Lecture 24	Chemoreception I: silkmoth pheromone plume-tracking (Kramer & Cardé)	
Wed Mar 24	Lecture 25	Chemoreception II: lobster olfactory search behavior	
Fri Mar 26	Lecture 26	Tactile specialization: the star-nosed mole's tactile fovea (Catania)	
Week 12	Lab 8	Lab 8 meets: rapid prey capture: star-nosed mole handling times and archerfish shot geometry from high-speed video	
Mon Mar 29	Lecture 27	Archerfish: predictive aiming and ballistic prey capture	Project proposal due 11:59 PM
Mon Mar 29	Lab 7	Lab 7 report due	Report due
Wed Mar 31	Lecture 28	Navigation I: desert ant path integration (Wehner)	
Fri Apr 2	Lecture 29	Navigation II: honeybee waggle dance and sun-compass navigation (von Frisch)	
Week 13	Lab 9	Lab 9 meets: orientation and navigation: circular statistics on monarch butterfly and desert ant headings	
Mon Apr 5	Lecture 30	Navigation III: monarch butterfly migration and the circadian sun compass	
Wed Apr 7	Lecture 31	Navigation IV: homing pigeon multimodal navigation	
Fri Apr 9	Lecture 32	Vocal communication I: songbird song-system anatomy (HVC, RA, Area X, LMAN)	
Sun Apr 11	Lab 8	Lab 8 report due	Report due
Week 14	Lab 10	Lab 10 meets: songbird song structure: syllable segmentation and song stereotypy in zebra finch recordings	
Mon Apr 12	Special Topic 3	Songbirds I: Artificial song memories, optogenetic tutoring, and neural mechanisms of vocal imitation	2–4 questions on Exam 4
Mon Apr 12	**Exam 3**	**Covers Lectures 21 to 30 (and 2–4 questions on Special Topic 2)**	**12:00 PM, NEU 100A or NEU 100B**
Wed Apr 14	Lecture 33	Vocal communication II: song learning, sensitive periods, and the template model (Marler & Konishi)	Project progress check due 11:59 PM
Thu Apr 15	Comp. 3	Computational Assignment 3: simulation: a leaky integrate-and-fire neuron	Due 11:59 PM

Date (2027)	Session	Topic	Assessment / Notes
Fri Apr 16	Lecture 34	Vocal communication III: Drosophila courtship song and the fruitless circuit	
Sun Apr 18	Lab 9	Lab 9 report due	Report due
Mon Apr 19	Lecture 35	Social behavior: crayfish dominance and serotonergic modulation of escape	
Wed Apr 21	Lecture 36	Simple learning circuits: Aplysia gill-withdrawal habituation and sensitization (Kandel)	
Fri Apr 23	Lecture 37	Spatial cognition: hippocampal place cells and grid cells (O'Keefe & Moser)	
Sat Apr 24	Reflection Essay	Captivity and Conservation Reflection (5 pages) due	Due 11:59 PM
Sun Apr 25	Lab 10	Lab 10 report due	Report due
Week 16	Make-up	Make-up lab for students who missed a scheduled lab	
Mon Apr 26	Lecture 38	Primate communication: vervet monkey alarm calls and referential signaling (Seyfarth & Cheney)	
Wed Apr 28	Lecture 39	Hormones and behavior: the ring dove reproductive cycle (Lehrman) and seasonal song plasticity	
Fri Apr 30	Lecture 40	Frontiers: optogenetics, wireless recording, and causal tests of neural circuits	
Fri May 7	Project	Lecture project final submission	Due 11:59 PM
Mon May 10	**Exam 4**	**Covers Lectures 31 to 40 (and 2–4 questions on Special Topic 3)**	**12:00 PM, NEU 100A or NEU 100B (final examination period)**

Readings Calendar

The readings below are recommended, not required. Each lecture lists the papers it draws on, taken from the reference lists in the lecture decks. Read them before lecture when you can, and use them to go deeper than the slides. I will teach the material in class.

Date (2027)	Session	Topic	Recommended Readings
Mon Jan 11	Lecture 1	Introduction to neuroethology: Tinbergen's four questions; proximate vs. ultimate causation	J.R. Barber; D. Plotkin; J.J. Rubin; N.T. Homziak; B.C. Leavell; P.R. Houlihan; K.A. Miner; J.W. Breinholt; B. Quirk-Royal; P.S. Padrón; M. Nunez; A.Y. Kawahara (2022). Anti-bat ultrasound production in moths is globally and phylogenetically widespread. Proceedings of the National Academy of Sciences 119(25):e2117485119. https://doi.org/10.1073/pnas.2117485119 M.J. Cormons; J. Zeil (2023). Digger wasps Microbembex monodonta SAY (Hymenoptera, Crabronidae) rely exclusively on visual cues when pinpointing their nest entrances. PLOS ONE 18(3):e0282144. https://doi.org/10.1371/journal.pone.0282144 N. Tinbergen (1963). On aims and methods of Ethology. Zeitschrift für Tierpsychologie 20:410–433. https://doi.org/10.1111/j.1439-0310.1963.tb01161.x O. Fehér; H. Wang; S. Saar; P.P. Mitra; O. Tchernichovski (2009). De novo establishment of wild-type song culture in the zebra finch. Nature 459(7246):564–568. https://doi.org/10.1038/nature07994
Wed Jan 13	Lecture 2	Fixed action patterns and sign stimuli: Lorenz and Tinbergen's stickleback and graylag goose work	K. Lorenz; N. Tinbergen (1938). Taxis und Instinkthandlung in der Eirollbewegung der Graugans. I. Zeitschrift für Tierpsychologie 2(1):1–29. Archive edition identifies 1938; Crossref indexes this DOI under 1939. https://doi.org/10.1111/j.1439-0310.1939.tb01558.x N. Tinbergen (1948). Social Releasers and the Experimental Method Required for Their Study. The

Date (2027)	Session	Topic	Recommended Readings
			Wilson Bulletin 60(1):6–51. Digitized article; repository DOI. https://doi.org/10.5281/zenodo.16185673 N. James; A. Bell (2021). Minimally invasive brain injections for viral-mediated transgenesis: New tools for behavioral genetics in sticklebacks. PLOS ONE 16(5):e0251653. https://doi.org/10.1371/journal.pone.0251653 S.A. Bukhari; M.C. Saul; C.H. Seward; H. Zhang; M. Bensky; N. James; S.D. Zhao; S. Chandrasekaran; L. Stubbs; A.M. Bell (2017). Temporal dynamics of neurogenomic plasticity in response to social interactions in male threespined sticklebacks. PLOS Genetics 13(7):e1006840. https://doi.org/10.1371/journal.pgen.1006840
Fri Jan 15	Lecture 3	Cockroach escape: the cercal system and giant interneurons (Camhi)	Booth, D., Marie, B., Domenici, P., Blagburn, J. M., & Bacon, J. P. (2009). Transcriptional control of behavior: engrailed knock-out changes cockroach escape trajectories. Journal of Neuroscience, 29(22), 7181–7190. https://doi.org/10.1523/JNEUROSCI.1374-09.2009 Camhi, J. M., Tom, W., & Volman, S. (1978). The escape behavior of the cockroach Periplaneta americana. II. Detection of natural predators by air displacement. Journal of Comparative Physiology, 128, 203–212. https://doi.org/10.1007/BF00656853 Camhi, J. M., & Tom, W. (1978). The escape behavior of the cockroach Periplaneta americana. I. Turning response to wind puffs. Journal of Comparative Physiology, 128, 193–201. https://doi.org/10.1007/BF00656852 Olsen, A. C. K., & Triblehorn, J. D. (2014). Neural responses from the filiform receptor neuron afferents of the wind-sensitive cercal system in three cockroach species. Journal of Insect Physiology, 68, 76–86. https://doi.org/10.1016/j.jinsphys.2014.07.002
Wed Jan 20	Lecture 4	Crayfish tail-flip: lateral and medial giant fiber circuits (Wine & Krasne)	J. Herberholz; S. H. Mishra; D. Uma; M. W. Germann; D. H. Edwards; K. Potter (2011). Non-invasive imaging of neuroanatomical structures and neural activation with high-resolution MRI. Frontiers in Behavioral Neuroscience 5:16. https://doi.org/10.3389/fnbeh.2011.00016 J. J. Wine; F. B. Krasne (1972). The Organization of Escape Behaviour in the Crayfish. Journal of Experimental Biology 56(1):1–18. https://doi.org/10.1242/jeb.56.1.1 M. E. Swierzbinski; J. Herberholz (2018). Effects of ethanol on sensory inputs to the medial giant interneurons of crayfish. Frontiers in Physiology 9:448. https://doi.org/10.3389/fphys.2018.00448 J. Herberholz; B. L. Antonsen; D. H. Edwards (2002). A lateral excitatory network in the escape circuit of crayfish. Journal of Neuroscience 22(20):9078–9085. https://doi.org/10.1523/JNEUROSCI.22-20-09078.2002
Fri Jan 22	Lecture 5	Fish startle: the Mauthner cell and the C-start response (Eaton & Faber)	Eaton RC, Bombardieri RA, Meyer DL (1977). The Mauthner-initiated startle response in teleost fish. Journal of Experimental Biology 66: 65–81. https://doi.org/10.1242/jeb.66.1.65 Eaton RC, DiDomenico R, Nissanov J (1988). Flexible body dynamics of the goldfish C-start: implications for reticulospinal command mechanisms. Journal of Neuroscience 8(8): 2758–2768. https://doi.org/10.1523/JNEUROSCI.08-08-02758.1988 Schuster S (2023). The archerfish predictive C-start. Journal of Comparative Physiology A 209(5): 827–837. https://doi.org/10.1007/s00359-023-01658-2 Burgess HA, Granato M (2007). Sensorimotor gating in larval zebrafish. Journal of Neuroscience 27(18): 4984–

Date (2027)	Session	Topic	Recommended Readings
			4994. https://doi.org/10.1523/JNEUROSCI.0615-07.2007
Mon Jan 25	Lecture 6	Central pattern generators I: leech swimming and crawling (Kristan)	M. Ashaber; Y. Tomina; P. Kassraian; E. A. Bushong; W. B. Kristan Jr; M. H. Ellisman; D. A. Wagenaar (2021). Anatomy and activity patterns in a multifunctional motor neuron and its surrounding circuits. eLife 10:e61881. https://doi.org/10.7554/eLife.61881 K. L. Briggman; W. B. Kristan Jr (2006). Imaging dedicated and multifunctional neural circuits generating distinct behaviors. Journal of Neuroscience 26(42):10925–10933. https://doi.org/10.1523/JNEUROSCI.3265-06.2006 T. W. Cacciatore; R. Rozenshteyn; W. B. Kristan Jr (2000). Kinematics and modeling of leech crawling: evidence for an oscillatory behavior produced by propagating waves of excitation. Journal of Neuroscience 20(4):1643–1655. https://doi.org/10.1523/JNEUROSCI.20-04-01643.2000 W. B. Kristan Jr; R. L. Calabrese (1976). Rhythmic swimming activity in neurones of the isolated nerve cord of the leech. Journal of Experimental Biology 65(3):643–668. https://doi.org/10.1242/jeb.65.3.643
Wed Jan 27	Lecture 7	Central pattern generators II: the lobster/crab stomatogastric ganglion (Marder & Selverston)	Powell D, Haddad SA, Gorur-Shandilya S, Marder E. (2021). Coupling between fast and slow oscillator circuits in Cancer borealis is temperature-compensated. eLife 10: e60454. https://doi.org/10.7554/elife.60454 Garcia VJ, Daur N, Temporal S, Schulz DJ, Bucher D. (2015). Neuropeptide receptor transcript expression levels and magnitude of ionic current responses show cell type-specific differences in a small motor circuit. Journal of Neuroscience 35: 6786-6800. https://doi.org/10.1523/jneurosci.0171-15.2015 Miller JP, Selverston AI. (1982). Mechanisms underlying pattern generation in lobster stomatogastric ganglion as determined by selective inactivation of identified neurons. II. Oscillatory properties of pyloric neurons. Journal of Neurophysiology 48: 1378-1391. https://doi.org/10.1152/jn.1982.48.6.1378 Hooper SL, Marder E. (1987). Modulation of the lobster pyloric rhythm by the peptide proctolin. Journal of Neuroscience 7: 2097-2112. https://doi.org/10.1523/jneurosci.07-07-02097.1987
Fri Jan 29	Lecture 8	Locust flight: wind-sensitive interneurons and the flight CPG	J. M. Camhi (1969c). Locust wind receptors. III. Contribution to flight initiation and lift control. Journal of Experimental Biology 50(2):363–373. https://doi.org/10.1242/jeb.50.2.363 D. M. Wilson (1961). The central nervous control of flight in a locust. Journal of Experimental Biology 38(2):471–490. https://doi.org/10.1242/jeb.38.2.471 E. Buhl; K. Schildberger; P. A. Stevenson (2008). A muscarinic cholinergic mechanism underlies activation of the central pattern generator for locust flight. Journal of Experimental Biology 211(14):2346–2357. https://doi.org/10.1242/jeb.017384 J. M. Camhi (1969a). Locust wind receptors. I. Transducer mechanics and sensory response. Journal of Experimental Biology 50(2):335–348. https://doi.org/10.1242/jeb.50.2.335
Mon Feb 1	Lecture 9	Amphibian prey capture I: toad visual prey-catching and tectal feature detectors (Ewert)	Ewert JP, Buxbaum-Conradi H, Dreisvogt F, Glagow M, Merkel-Harff C, Röttgen A, Schürg-Pfeiffer E, Schwippert WW (2001). Neural modulation of visuomotor functions underlying prey-catching behaviour in anurans: perception, attention, motor performance, learning. Comparative Biochemistry and Physiology A 128: 417–461. https://doi.org/10.1016/S1095-6433(00)00333-0 Ewert JP, Arend B, Becker V, Borchers HW (1979). Invariants in configurational prey selection by Bufo bufo

Date (2027)	Session	Topic	Recommended Readings
			(L.). Brain, Behavior and Evolution 16: 38–51. https://doi.org/10.1159/000121822 Ewert JP (1997). Neural correlates of key stimulus and releasing mechanism: a case study and two concepts. Trends in Neurosciences 20(8): 332–339. https://doi.org/10.1016/S0166-2236(96)01042-9 Schürg-Pfeiffer E, Ewert JP (1981). Investigation of neurons involved in the analysis of gestalt prey features in the frog Rana temporaria. Journal of Comparative Physiology 141: 139–152. https://doi.org/10.1007/BF01342660
Wed Feb 3	Lecture 10	Amphibian prey capture II: pretectal predator-avoidance circuits	Ewert JP (1970). Neural mechanisms of prey-catching and avoidance behavior in the toad (Bufo bufo L.). Brain, Behavior and Evolution 3(1–4):36–56. https://doi.org/10.1159/000125462 Ingle D (1973). Disinhibition of tectal neurons by pretectal lesions in the frog. Science 180(4084):422–424. https://doi.org/10.1126/science.180.4084.422 Laming PR, Ewert JP (1983). The effects of pretectal lesions on neuronal, sustained potential shift and electroencephalographic responses of the toad tectum to presentation of a visual stimulus. Comparative Biochemistry and Physiology Part A: Physiology 76(2):247–252. https://doi.org/10.1016/0300-9629(83)90322-5 Schwippert WW, Beneke TW, Ewert JP (1995). Pretecto-tectal influences. II. How retinal and pretectal inputs to the toad's superficial tectum interact: a study of electrically evoked field potentials. Journal of Comparative Physiology A 176(2):181–192. https://doi.org/10.1007/BF00239921
Fri Feb 5	Lecture 11	Insect vision I: the fly optomotor response and the Reichardt elementary motion detector	Mano O, Choi M, Tanaka R, Creamer MS, Matos NCB, Shomar JW, Badwan BA, Clandinin TR, Clark DA (2023). Long-timescale anti-directional rotation in Drosophila optomotor behavior. eLife 12:e86076. https://doi.org/10.7554/elife.86076 Duistermars BJ, Chow DM, Condro M, Frye MA (2007). The spatial, temporal and contrast properties of expansion and rotation flight optomotor responses in Drosophila. Journal of Experimental Biology 210:3218-3227. https://doi.org/10.1242/jeb.007807 Hassenstein B, Reichardt W (1956). Systemtheoretische Analyse der Zeit-, Reihenfolgen- und Vorzeichenauswertung bei der Bewegungsperzeption des Rüsselkäfers Chlorophanus. Zeitschrift für Naturforschung B 11:513–524. https://doi.org/10.1515/znb-1956-9-1004 Clark DA, Bursztyn L, Horowitz MA, Schnitzer MJ, Clandinin TR (2011). Defining the computational structure of the motion detector in Drosophila. Neuron 70:1165-1177. https://doi.org/10.1016/j.neuron.2011.05.023
Mon Feb 8	Special Topic 1	Jellyfish I: neuroanatomy of Cassiopea, distributed nerve nets, rhopalia, and pacemaker control of rhythmic bell contractions	Amplatz K, Zieger E, Abed-Navandi D, Weissenbacher A, Wanninger A (2024). Neuromuscular development in the emerging scyphozoan model system, Cassiopea xamachana: implications for the evolution of cnidarian nervous systems. Frontiers in Neuroscience 17: 1324980. https://doi.org/10.3389/fnins.2023.1324980 Nakanishi N, Hartenstein V, Jacobs DK (2009). Development of the rhopalial nervous system in Aurelia sp.1 (Cnidaria, Scyphozoa). Development Genes and Evolution 219(6): 301-317. https://doi.org/10.1007/s00427-009-0291-y Muffett KM, Mammone M, Puckett J, Martino G, Aditi, Miglietta MP (2025). Regeneration and Musculature in Halved Cassiopea xamachana Ephyrae. Integrative

Date (2027)	Session	Topic	Recommended Readings
			Organismal Biology 7(1): obaf030. https://doi.org/10.1093/iob/obaf030 Anuszczyk SR, Yoder N, Costello JH, Dabiri JO, Gemmell BJ, Rutledge KM, Colin SP (2025). Increasing the Reliability and Versatility of Jellyfish Biohybrid Vehicles via Species Selection and Rhopalia Removal. Biomimetics 10(12): 810. https://doi.org/10.3390/biomimetics10120810
Wed Feb 10	Lecture 12	Insect vision II: jumping spider eyes and predatory tracking (Land)	Zurek DB, Cronin TW, Taylor LA, Byrne K, Sullivan MLG, Morehouse NI (2015). Spectral filtering enables trichromatic vision in colorful jumping spiders. Current Biology 25: R403–R404. https://doi.org/10.1016/j.cub.2015.03.033 Zurek DB, Taylor AJ, Evans CS, Nelson XJ (2010). The role of the anterior lateral eyes in the vision-based behaviour of jumping spiders. Journal of Experimental Biology 213: 2372–2378. https://doi.org/10.1242/jeb.042382 Land MF (1969a). Structure of the retinae of the principal eyes of jumping spiders (Salticidae: Dendryphantinae) in relation to visual optics. Journal of Experimental Biology 51: 443–470. https://doi.org/10.1242/jeb.51.2.443 Nagata T, Koyanagi M, Tsukamoto H, Saeki S, Isono K, Shichida Y, Tokunaga F, Kinoshita M, Arikawa K, Terakita A (2012). Depth perception from image defocus in a jumping spider. Science 335: 469–471. https://doi.org/10.1126/science.1211667
Fri Feb 12	Lecture 13	Mantis shrimp vision: hyperspectral and polarization sensing	Bok MJ, Roberts NW, Cronin TW (2018). Behavioural evidence for polychromatic ultraviolet sensitivity in mantis shrimp. Proceedings of the Royal Society B: Biological Sciences 285: 20181384. https://doi.org/10.1098/rspb.2018.1384 Thoen HH, How MJ, Chiou TH, Marshall J (2014). A Different Form of Color Vision in Mantis Shrimp. Science 343: 411-413. https://doi.org/10.1126/science.1245824 Wang CWJ, Marshall NJ (2025). Behavioural evidence of spectral opponent processing in the visual system of stomatopod crustaceans. Journal of Experimental Biology 228: jeb247952. https://doi.org/10.1242/jeb.247952 Kleinlogel S, White AG (2008). The Secret World of Shrimps: Polarisation Vision at Its Best. PLoS ONE 3: e2190. https://doi.org/10.1371/journal.pone.0002190
Wed Feb 17	Lecture 14	Insect hearing I: cricket phonotaxis and song recognition (Hoy)	Pollack GS, Hoy RR (1979). Temporal pattern as a cue for species-specific calling song recognition in crickets. Science 204(4391):429–432. https://doi.org/10.1126/science.204.4391.429 Kostarakos K, Hedwig B (2012). Calling song recognition in female crickets: temporal tuning of identified brain neurons matches behavior. Journal of Neuroscience 32(28):9601–9612. https://doi.org/10.1523/JNEUROSCI.1170-12.2012 Hoy RR, Paul RC (1973). Genetic control of song specificity in crickets. Science 180(4081):82–83. https://doi.org/10.1126/science.180.4081.82 Michelsen A, Popov AV, Lewis B (1994). Physics of directional hearing in the cricket Gryllus bimaculatus. Journal of Comparative Physiology A 175:153–164. https://doi.org/10.1007/BF00215111
Fri Feb 19	Lecture 15	Insect hearing II: moth ears and bat-evasion behavior (Roeder)	Moir HM, Jackson JC, Windmill JFC (2013). Extremely high frequency sensitivity in a ‘simple’ ear. Biology Letters 9: 20130241. https://doi.org/10.1098/rsbl.2013.0241 Roeder KD (1962). The behaviour of free flying moths in the presence of artificial ultrasonic pulses. Animal Behaviour 10: 300-304. https://doi.org/10.1016/0003-

Date (2027)	Session	Topic	Recommended Readings
			3472(62)90053-2 Corcoran AJ, Conner WE (2016). How moths escape bats: predicting outcomes of predator-prey interactions. Journal of Experimental Biology 219: 2704–2715. https://doi.org/10.1242/jeb.137638 Roeder KD, Treat AE (1957). Ultrasonic reception by the tympanic organ of noctuid moths. Journal of Experimental Zoology 134: 127-157. https://doi.org/10.1002/jez.1401340107
Mon Feb 22	Lecture 16	Bat echolocation I: CF-FM bats and Doppler-shift compensation (Suga)	Schoeppler D, Schnitzler H-U, Denzinger A (2018). Precise Doppler shift compensation in the hipposiderid bat, Hipposideros armiger. Scientific Reports 8:4598. https://doi.org/10.1038/s41598-018-22880-y Suga N, Simmons JA, Jen PH-S (1975). Peripheral specialization for fine analysis of Doppler-shifted echoes in the auditory system of the “CF-FM” bat Pteronotus parnellii. Journal of Experimental Biology 63(1):161–192. https://doi.org/10.1242/jeb.63.1.161 Schuller G, Beuter K, Schnitzler H-U (1974). Response to frequency shifted artificial echoes in the bat Rhinolophus ferrumequinum. Journal of Comparative Physiology 89(3):275–286. https://doi.org/10.1007/BF00696191 Metzner W, Zhang S, Smotherman M (2002). Doppler-shift compensation behavior in horseshoe bats revisited: auditory feedback controls both a decrease and an increase in call frequency. Journal of Experimental Biology 205(11):1607–1616. https://doi.org/10.1242/jeb.205.11.1607
Wed Feb 24	Lecture 17	Bat echolocation II: FM bats, target-range computation, clutter rejection	Bates ME, Simmons JA, Zorikov TV (2011). Bats use echo harmonic structure to distinguish their targets from background clutter. Science 333(6042):627–630. https://doi.org/10.1126/science.1202065 Ming C, Bates ME, Simmons JA (2020). How frequency hopping suppresses pulse-echo ambiguity in bat biosonar. Proceedings of the National Academy of Sciences 117(29):17288–17295. https://doi.org/10.1073/pnas.2001105117 Simmons JA (1973). The resolution of target range by echolocating bats. Journal of the Acoustical Society of America 54(1):157–173. https://doi.org/10.1121/1.1913559 Simmons AM, Warnecke M, Simmons JA (2024). Microseconds-level coding of echo delay in the auditory brainstem of an FM-echolocating bat. Journal of Neurophysiology 132(6):2012–2022. https://doi.org/10.1152/jn.00305.2024
Fri Feb 26	Lecture 18	Owl sound localization I: ITD/ILD maps in the inferior colliculus (Konishi & Knudsen)	Knudsen EI, Konishi M (1978). A Neural Map of Auditory Space in the Owl. Science 200(4343):795-797. https://doi.org/10.1126/science.644324 Takahashi T, Moiseff A, Konishi M (1984). Time and intensity cues are processed independently in the auditory system of the owl. The Journal of Neuroscience 4(7):1781-1786. https://doi.org/10.1523/jneurosci.04-07-01781.1984 Carr CE, Konishi M (1990). A circuit for detection of interaural time differences in the brain stem of the barn owl. The Journal of Neuroscience 10(10):3227-3246. https://doi.org/10.1523/jneurosci.10-10-03227.1990 Sullivan WE, Konishi M (1984). Segregation of stimulus phase and intensity coding in the cochlear nucleus of the barn owl. The Journal of Neuroscience 4(7):1787-1799. https://doi.org/10.1523/jneurosci.04-07-01787.1984
Mon Mar 1	Lecture 19	Owl sound localization II: developmental plasticity of the auditory map	MS Brainard; EI Knudsen (1993). Experience-dependent plasticity in the inferior colliculus: a site for visual calibration of the neural representation of auditory space

Date (2027)	Session	Topic	Recommended Readings
			in the barn owl. The Journal of Neuroscience 13(11):4589-4608. https://doi.org/10.1523/jneurosci.13-11-04589.1993 Daniel E. Feldman; Eric I. Knudsen (1998). Pharmacological Specialization of Learned Auditory Responses in the Inferior Colliculus of the Barn Owl. The Journal of Neuroscience 18(8):3073-3087. https://doi.org/10.1523/jneurosci.18-08-03073.1998 Peter S. Hyde; Eric I. Knudsen (2001). A Topographic Instructive Signal Guides the Adjustment of the Auditory Space Map in the Optic Tectum. The Journal of Neuroscience 21(21):8586-8593. https://doi.org/10.1523/jneurosci.21-21-08586.2001 Michael S. Brainard; Eric I. Knudsen (1998). Sensitive Periods for Visual Calibration of the Auditory Space Map in the Barn Owl Optic Tectum. The Journal of Neuroscience 18(10):3929-3942. https://doi.org/10.1523/jneurosci.18-10-03929.1998
Wed Mar 3	Lecture 20	Electroreception I: passive sensing in sharks and rays (ampullae of Lorenzini)	Bellono NW, Leitch DB, Julius D (2017). Molecular basis of ancestral vertebrate electroreception. Nature 543(7645):391-396. https://doi.org/10.1038/nature21401 Kalmijn AJ (1971). The Electric Sense of Sharks and Rays. Journal of Experimental Biology 55(2):371-383. https://doi.org/10.1242/jeb.55.2.371 Kajiura SM, Holland KN (2002). Electroreception in juvenile scalloped hammerhead and sandbar sharks. Journal of Experimental Biology 205(23):3609-3621. https://doi.org/10.1242/jeb.205.23.3609 Josberger EE, Hassanzadeh P, Deng Y, Sohn J, Rego MJ, Amemiya CT, Rolandi M (2016). Proton conductivity in ampullae of Lorenzini jelly. Science Advances 2(5):e1600112. https://doi.org/10.1126/sciadv.1600112
Fri Mar 5	Lecture 21	Electroreception II: active electrolocation in weakly electric fish (Bullock)	G. von der Emde; K. Behr; B. Bouton; J. Engelmann; S. Fetz; C. Folde (2010). 3-Dimensional scene perception during active electrolocation in a weakly electric pulse fish. Frontiers in Behavioral Neuroscience 4:26. https://doi.org/10.3389/fnbeh.2010.00026 V. Hofmann; J.I. Sanguinetti-Scheck; L. Gómez-Sena; J. Engelmann (2017). Sensory Flow as a Basis for a Novel Distance Cue in Freely Behaving Electric Fish. Journal of Neuroscience 37(2):302–312. https://doi.org/10.1523/JNEUROSCI.1361-16.2016 S. Schumacher; T. Burt de Perera; G. von der Emde (2017). Electrosensory capture during multisensory discrimination of nearby objects in the weakly electric fish Gnathonemus petersii. Scientific Reports 7:43665. https://doi.org/10.1038/srep43665 N.B. Sawtell; A. Williams; P.D. Roberts; G. von der Emde; C.C. Bell (2006). Effects of Sensing Behavior on a Latency Code. Journal of Neuroscience 26(32):8221–8234. https://doi.org/10.1523/JNEUROSCI.1508-06.2006
Mon Mar 8	Special Topic 2	Axolotl: lateral-line sensing, electroreception, and sensory processing of courtship signals	Wang Y, Pan Y, Liu X, Zhang X, Fan C, Wang J (2026). Characterization of the electrosensory receptor in axolotl and the role of voltage-gated potassium channels in electrosensitivity. Comparative Biochemistry and Physiology Part A 319:112051. https://doi.org/10.1016/j.cbpa.2026.112051 Münz H, Claas B, Fritzsch B (1982). Electrophysiological evidence of electroreception in the axolotl Siredon mexicanum. Neuroscience Letters 28:107–111. https://doi.org/10.1016/0304-3940(82)90216-6 Bartels M, Münz H, Claas B (1990). Representation of lateral line and electrosensory systems in the midbrain of the axolotl, Ambystoma mexicanum. Journal of Comparative Physiology A 167:347–356. https://doi.org/10.1007/BF00192570

Date (2027)	Session	Topic	Recommended Readings
			Maex M, Van Bocxlaer I, Mortier A, Proost P, Bossuyt F (2016). Courtship Pheromone Use in a Model Urodele, the Mexican Axolotl (Ambystoma mexicanum). Scientific Reports 6:20184. https://doi.org/10.1038/srep20184
Wed Mar 10	Lecture 22	Electroreception III: the jamming avoidance response (Heiligenberg)	Carr C, Heiligenberg W, Rose G (1986). A time-comparison circuit in the electric fish midbrain. I. Behavior and physiology. The Journal of Neuroscience 6(1):107-119. https://doi.org/10.1523/jneurosci.06-01-00107.1986 Rose G, Heiligenberg W (1985). Temporal hyperacuity in the electric sense of fish. Nature 318(6042):178-180. https://doi.org/10.1038/318178a0 Carr C, Maler L, Taylor B (1986). A time-comparison circuit in the electric fish midbrain. II. Functional morphology. The Journal of Neuroscience 6(5):1372-1383. https://doi.org/10.1523/jneurosci.06-05-01372.1986 Heiligenberg W, Rose G (1985). Phase and amplitude computations in the midbrain of an electric fish: intracellular studies of neurons participating in the jamming avoidance response of Eigenmannia. The Journal of Neuroscience 5(2):515-531. https://doi.org/10.1523/jneurosci.05-02-00515.1985
Fri Mar 12	Lecture 23	Magnetoreception: sea turtle and migratory bird magnetic compasses	Lohmann KJ (1991). Magnetic Orientation By Hatchling Loggerhead Sea Turtles (Caretta Caretta). Journal of Experimental Biology 155(1):37-49. https://doi.org/10.1242/jeb.155.1.37 Lohmann KJ, Lohmann CMF (1994). Detection of Magnetic Inclination Angle by Sea Turtles: A Possible Mechanism for Determining Latitude. Journal of Experimental Biology 194(1):23-32. https://doi.org/10.1242/jeb.194.1.23 Lohmann KJ, Cain SD, Dodge SA, Lohmann CMF (2001). Regional Magnetic Fields as Navigational Markers for Sea Turtles. Science 294(5541):364-366. https://doi.org/10.1126/science.1064557 Goforth KM, Lohmann CMF, Gavin A, Henning R, Harvey A, Hinton TL, Lim DS, Lohmann KJ (2025). Learned magnetic map cues and two mechanisms of magnetoreception in turtles. Nature 638(8052):1015-1022. https://doi.org/10.1038/s41586-024-08554-y
Mon Mar 22	Lecture 24	Chemoreception I: silkmoth pheromone plume-tracking (Kramer & Cardé)	Yamada M, Ohashi H, Hosoda K, Kurabayashi D, Shigaki S (2021). Multisensory-motor integration in olfactory navigation of silkmoth, Bombyx mori, using virtual reality system. eLife 10:e72001. https://doi.org/10.7554/elife.72001 Shiota Y, Sakurai T, Ando N, Haupt SS, Mitsuno H, Daimon T, Kanzaki R (2021). Pheromone binding protein is involved in temporal olfactory resolution in the silkmoth. iScience 24(11):103334. https://doi.org/10.1016/j.isci.2021.103334 Vickers NJ, Baker TC (1994). Reiterative responses to single strands of odor promote sustained upwind flight and odor source location by moths. Proceedings of the National Academy of Sciences 91(13):5756-5760. https://doi.org/10.1073/pnas.91.13.5756 Sakurai T, Mitsuno H, Haupt SS, Uchino K, Yokohari F, Nishioka T, Kobayashi I, Sezutsu H, Tamura T, Kanzaki R (2011). A Single Sex Pheromone Receptor Determines Chemical Response Specificity of Sexual Behavior in the Silkmoth Bombyx mori. PLoS Genetics 7(6):e1002115. https://doi.org/10.1371/journal.pgen.1002115
Wed Mar 24	Lecture 25	Chemoreception II: lobster olfactory search behavior	Moore PA, Scholz N, Atema J (1991). Chemical orientation of lobsters, homarus americanus, in turbulent odor plumes. Journal of Chemical Ecology 17(7):1293-

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			1307. https://doi.org/10.1007/bf00983763 Horner AJ, Weissburg MJ, Derby CD (2004). Dual antennular chemosensory pathways can mediate orientation by Caribbean spiny lobsters in naturalistic flow conditions. Journal of Experimental Biology 207(21):3785-3796. https://doi.org/10.1242/jeb.01200 Reidenbach MA, Koehl MAR (2011). The spatial and temporal patterns of odors sampled by lobsters and crabs in a turbulent plume. Journal of Experimental Biology 214(18):3138–3153. https://doi.org/10.1242/jeb.057547 Reidenbach MA, George NT, Koehl MAR (2008). Antennule morphology and flicking kinematics facilitate odor sampling by the spiny lobster, Panulirus argus. Journal of Experimental Biology 211(17):2849-2858. https://doi.org/10.1242/jeb.016394
Fri Mar 26	Lecture 26	Tactile specialization: the star-nosed mole's tactile fovea (Catania)	Gerhold KA, Pellegrino M, Tsunozaki M, Morita T, Leitch DB, Tsuruda PR, Brem RB, Catania KC, Bautista DM (2013). The Star-Nosed Mole Reveals Clues to the Molecular Basis of Mammalian Touch. PLoS ONE 8(1):e55001. https://doi.org/10.1371/journal.pone.0055001 Catania KC, Kaas JH (1997). Somatosensory fovea in the star-nosed mole: Behavioral use of the star in relation to innervation patterns and cortical representation. The Journal of Comparative Neurology 387(2):215-233. https://doi.org/10.1002/(sici)1096-9861(19971020)387:2<215::aid-cne4>3.0.co;2-3 Catania KC, Remple FE (2005). Asymptotic prey profitability drives star-nosed moles to the foraging speed limit. Nature 433(7025):519-522. https://doi.org/10.1038/nature03250 Sawyer EK, Catania KC (2016). Somatosensory organ topography across the star of the star-nosed mole (Condylura cristata). Journal of Comparative Neurology 524(5):917-929. https://doi.org/10.1002/cne.23943
Mon Mar 29	Lecture 27	Archerfish: predictive aiming and ballistic prey capture	Reichenthal A, Ben-Tov M, Segev R (2018). Coding Schemes in the Archerfish Optic Tectum. Frontiers in Neural Circuits 12: 18. https://doi.org/10.3389/fncir.2018.00018 Rossel S, Corlija J, Schuster S (2002). Predicting three-dimensional target motion: how archer fish determine where to catch their dislodged prey. Journal of Experimental Biology 205: 3321-3326. https://doi.org/10.1242/jeb.205.21.3321 Gerullis P, Schuster S (2014). Archerfish Actively Control the Hydrodynamics of Their Jets. Current Biology 24: 2156-2160. https://doi.org/10.1016/j.cub.2014.07.059 Temple S, Hart NS, Marshall NJ, Collin SP (2010). A spitting image: specializations in archerfish eyes for vision at the interface between air and water. Proceedings of the Royal Society B: Biological Sciences 277: 2607-2615. https://doi.org/10.1098/rspb.2010.0345
Wed Mar 31	Lecture 28	Navigation I: desert ant path integration (Wehner)	Wittlinger M, Wehner R, Wolf H (2006). The Ant Odometer: Stepping on Stilts and Stumps. Science 312(5782):1965–1967. https://doi.org/10.1126/science.1126912 Wittlinger M, Wehner R, Wolf H (2007). The desert ant odometer: a stride integrator that accounts for stride length and walking speed. Journal of Experimental Biology 210(2):198–207. https://doi.org/10.1242/jeb.02657 Grob R, Holland Cunz O, Grübel K, Pfeiffer K, Rössler W, Fleischmann PN (2022). Rotation of skylight polarization during learning walks is necessary to trigger neuronal plasticity in Cataglyphis ants. Proceedings of

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			the Royal Society B 289:20212499. https://doi.org/10.1098/rspb.2021.2499 Grob R, Müller VL, Grübel K, Rössler W, Fleischmann PN (2024). Importance of magnetic information for neuronal plasticity in desert ants. Proceedings of the National Academy of Sciences 121(8):e2320764121. https://doi.org/10.1073/pnas.2320764121
Fri Apr 2	Lecture 29	Navigation II: honeybee waggle dance and sun-compass navigation (von Frisch)	Dong S, Lin T, Nieh JC, Tan K (2023). Social signal learning of the waggle dance in honey bees. Science 379(6636): 1015-1018. https://doi.org/10.1126/science.ade1702 Riley JR, Greggers U, Smith AD, Reynolds DR, Menzel R (2005). The flight paths of honeybees recruited by the waggle dance. Nature 435(7039): 205-207. https://doi.org/10.1038/nature03526 Srinivasan MV, Zhang S, Altwein M, Tautz J (2000). Honeybee Navigation: Nature and Calibration of the "Odometer". Science 287(5454): 851-853. https://doi.org/10.1126/science.287.5454.851 Esch HE, Zhang S, Srinivasan MV, Tautz J (2001). Honeybee dances communicate distances measured by optic flow. Nature 411(6837): 581-583. https://doi.org/10.1038/35079072
Mon Apr 5	Lecture 30	Navigation III: monarch butterfly migration and the circadian sun compass	Henrik Mouritsen; Barrie J. Frost (2002). Virtual migration in tethered flying monarch butterflies reveals their orientation mechanisms. Proceedings of the National Academy of Sciences, 99(15): 10162-10166. https://doi.org/10.1073/pnas.152137299 Oren Froy; Anthony L. Gotter; Amy L. Casselman; Steven M. Reppert (2003). Illuminating the Circadian Clock in Monarch Butterfly Migration. Science, 300(5623): 1303-1305. https://doi.org/10.1126/science.1084874 Ivo Sauman; Adriana D. Briscoe; Haisun Zhu; Dingding Shi; Oren Froy; Julia Stalleicken; Quan Yuan; Amy Casselman; Steven M. Reppert (2005). Connecting the Navigational Clock to Sun Compass Input in Monarch Butterfly Brain. Neuron, 46(3): 457-467. https://doi.org/10.1016/j.neuron.2005.03.014 Julia Stalleicken; Maya Mukhida; Thomas Labhart; Rüdiger Wehner; Barrie Frost; Henrik Mouritsen (2005). Do monarch butterflies use polarized skylight for migratory orientation? Journal of Experimental Biology, 208(12): 2399-2408. https://doi.org/10.1242/jeb.01613
Wed Apr 7	Lecture 31	Navigation IV: homing pigeon multimodal navigation	Biro D, Meade J, Guilford T (2004). Familiar route loyalty implies visual pilotage in the homing pigeon. Proceedings of the National Academy of Sciences 101(50):17440–17443. https://doi.org/10.1073/pnas.0406984101 Biro D, Freeman R, Meade J, Roberts S, Guilford T (2007). Pigeons combine compass and landmark guidance in familiar route navigation. Proceedings of the National Academy of Sciences 104(18):7471–7476. https://doi.org/10.1073/pnas.0701575104 Gagliardo A, Ioalè P, Filannino C, Wikelski M (2011). Homing pigeons only navigate in air with intact environmental odours: a test of the olfactory activation hypothesis with GPS data loggers. PLOS ONE 6(8):e22385. https://doi.org/10.1371/journal.pone.0022385 Jorge PE, Phillips JB, Gonçalves A, Marques PAM, Němec P (2014). Odours stimulate neuronal activity in the dorsolateral area of the hippocampal formation during path integration. Proceedings of the Royal Society B 281(1783):20140025. https://doi.org/10.1098/rspb.2014.0025

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Fri Apr 9	Lecture 32	Vocal communication I: songbird song-system anatomy (HVC, RA, Area X, LMAN)	Nottebohm F, Stokes TM, Leonard CM (1976). Central control of song in the canary, Serinus canarius. Journal of Comparative Neurology 165(4):457-486. https://doi.org/10.1002/cne.901650405 Bottjer SW, Miesner EA, Arnold AP (1984). Forebrain Lesions Disrupt Development But Not Maintenance of Song in Passerine Birds. Science 224(4651):901-903. https://doi.org/10.1126/science.6719123 Benezra SE, Narayanan RT, Egger R, Oberlaender M, Long MA (2018). Morphological characterization of HVC projection neurons in the zebra finch (Taeniopygia guttata). Journal of Comparative Neurology 526(10):1673-1689. https://doi.org/10.1002/cne.24437 Doupe AJ (1997). Song- and Order-Selective Neurons in the Songbird Anterior Forebrain and their Emergence during Vocal Development. The Journal of Neuroscience 17(3):1147-1167. https://doi.org/10.1523/jneurosci.17-03-01147.1997
Mon Apr 12	Special Topic 3	Songbirds I: Artificial song memories, optogenetic tutoring, and neural mechanisms of vocal imitation	Zhao W, Garcia-Oscos F, Dinh D, Roberts TF (2019). Inception of memories that guide vocal learning in the songbird. Science 366:83–89. https://doi.org/10.1126/science.aaw4226 Roberts TF, Gobes SMH, Murugan M, Ölveczky BP, Mooney R (2012). Motor circuits are required to encode a sensory model for imitative learning. Nature Neuroscience 15:1454–1459. https://doi.org/10.1038/nn.3206 Tanaka M, Sun F, Li Y, Mooney R (2018). A mesocortical dopamine circuit enables the cultural transmission of vocal behaviour. Nature 563:117–120. https://doi.org/10.1038/s41586-018-0636-7 Roberts TF, Tschida KA, Klein ME, Mooney R (2010). Rapid spine stabilization and synaptic enhancement at the onset of behavioural learning. Nature 463:948–952. https://doi.org/10.1038/nature08759
Wed Apr 14	Lecture 33	Vocal communication II: song learning, sensitive periods, and the template model (Marler & Konishi)	Marler P (1970). A comparative approach to vocal learning: Song development in white-crowned sparrows. Journal of Comparative and Physiological Psychology 71(2, Pt. 2): 1–25. https://doi.org/10.1037/h0029144 Brainard MS, Doupe AJ (2000). Auditory feedback in learning and maintenance of vocal behaviour. Nature Reviews Neuroscience 1: 31–40. https://doi.org/10.1038/35036205 Marler P, Peters S (1981). Sparrows learn adult song and more from memory. Science 213(4509): 780–782. https://doi.org/10.1126/science.213.4509.780 Margoliash D, Konishi M (1985). Auditory representation of autogenous song in the song system of white-crowned sparrows. Proceedings of the National Academy of Sciences USA 82(17): 5997–6000. https://doi.org/10.1073/pnas.82.17.5997
Fri Apr 16	Lecture 34	Vocal communication III: Drosophila courtship song and the fruitless circuit	Coen P, Clemens J, Weinstein AJ, Pacheco DA, Deng Y, Murthy M (2014). Dynamic sensory cues shape song structure in Drosophila. Nature 507(7491):233–237. https://doi.org/10.1038/nature13131 Demir E, Dickson BJ (2005). fruitless splicing specifies male courtship behavior in Drosophila. Cell 121(5):785–794. https://doi.org/10.1016/j.cell.2005.04.027 Amin H, Nolte SS, Swain B, von Philipsborn AC (2023). GABAergic signaling shapes multiple aspects of Drosophila courtship motor behavior. iScience 26(11):108069. https://doi.org/10.1016/j.isci.2023.108069 Roemschied FA, Pacheco DA, Aragon MJ, Ireland EC, Li X, Thieringer K, Pang R, Murthy M (2023). Flexible circuit mechanisms for context-dependent song

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			sequencing. Nature 622(7984):794–801. https://doi.org/10.1038/s41586-023-06632-1
Mon Apr 19	Lecture 35	Social behavior: crayfish dominance and serotonergic modulation of escape	Jens Herberholz; Subrata H. Mishra; Divya Uma; Markus W. Germann; Donald H. Edwards; Kimberlee Potter (2011). Non-Invasive Imaging of Neuroanatomical Structures and Neural Activation with High-Resolution MRI. Frontiers in Behavioral Neuroscience 5:16. https://doi.org/10.3389/fnbeh.2011.00016 Jens Herberholz; Fadi A. Issa; Donald H. Edwards (2001). Patterns of Neural Circuit Activation and Behavior during Dominance Hierarchy Formation in Freely Behaving Crayfish. The Journal of Neuroscience, 21(8): 2759-2767. https://doi.org/10.1523/jneurosci.21-08-02759.2001 Robert Huber; Kalim Smith; Antonia Delago; Karin Isaksson; Edward A. Kravitz (1997). Serotonin and aggressive motivation in crustaceans: Altering the decision to retreat. Proceedings of the National Academy of Sciences, 94(11): 5939-5942. https://doi.org/10.1073/pnas.94.11.5939 Shih-Rung Yeh; Barbara E. Musolf; Donald H. Edwards (1997). Neuronal Adaptations to Changes in the Social Dominance Status of Crayfish. The Journal of Neuroscience, 17(2): 697-708. https://doi.org/10.1523/jneurosci.17-02-00697.1997
Wed Apr 21	Lecture 36	Simple learning circuits: Aplysia gill-withdrawal habituation and sensitization (Kandel)	Pinsker H; Kupfermann I; Castellucci V; Kandel E (1970). Habituation and Dishabituation of the Gill-Withdrawal Reflex in Aplysia. Science, 167(3926): 1740-1742. https://doi.org/10.1126/science.167.3926.1740 Castellucci V; Pinsker H; Kupfermann I; Kandel ER (1970). Neuronal Mechanisms of Habituation and Dishabituation of the Gill-Withdrawal Reflex in Aplysia. Science, 167(3926): 1745-1748. https://doi.org/10.1126/science.167.3926.1745 Brunelli M; Castellucci V; Kandel ER (1976). Synaptic Facilitation and Behavioral Sensitization in Aplysia : Possible Role of Serotonin and Cyclic AMP. Science, 194(4270): 1178-1181. https://doi.org/10.1126/science.186870 Montarolo PG; Goelet P; Castellucci VF; Morgan J; Kandel ER; Schacher S (1986). A Critical Period for Macromolecular Synthesis in Long-Term Heterosynaptic Facilitation in Aplysia. Science, 234(4781): 1249-1254. https://doi.org/10.1126/science.3775383
Fri Apr 23	Lecture 37	Spatial cognition: hippocampal place cells and grid cells (O'Keefe & Moser)	O'Keefe J, Dostrovsky J (1971). The hippocampus as a spatial map. Preliminary evidence from unit activity in the freely-moving rat. Brain Research 34(1): 171–175. https://doi.org/10.1016/0006-8993(71)90358-1 Giocomo LM, Hussaini SA, Zheng F, Kandel ER, Moser MB, Moser EI (2011). Grid cells use HCN1 channels for spatial scaling. Cell 147(5): 1159–1170. https://doi.org/10.1016/j.cell.2011.08.051 Hafting T, Fyhn M, Molden S, Moser MB, Moser EI (2005). Microstructure of a spatial map in the entorhinal cortex. Nature 436(7052): 801–806. https://doi.org/10.1038/nature03721
Mon Apr 26	Lecture 38	Primate communication: vervet monkey alarm calls and referential signaling (Seyfarth & Cheney)	Robert M. Seyfarth, Dorothy L. Cheney, Peter Marler (1980). Monkey Responses to Three Different Alarm Calls: Evidence of Predator Classification and Semantic Communication. Science 210(4471):801-803. https://doi.org/10.1126/science.7433999 Robert M. Seyfarth, Dorothy L. Cheney, Peter Marler (1980). Vervet monkey alarm calls: Semantic communication in a free-ranging primate. Animal Behaviour 28(4):1070-1094. https://doi.org/10.1016/S0003-3472(80)80097-2

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			Tabitha Price, Philip Wadewitz, Dorothy Cheney, Robert Seyfarth, Kurt Hammerschmidt, Julia Fischer (2015). Vervets revisited: A quantitative analysis of alarm call structure and context specificity. Scientific Reports 5(1):13220. https://doi.org/10.1038/srep13220 Michael Ortiz-Rios, Paweł Kuśmierek, Iain DeWitt, Denis Archakov, Frederico A. C. Azevedo, Mikko Sams, Iiro P. Jääskeläinen, Georgios A. Keliris, Josef P. Rauschecker (2015). Functional MRI of the vocalization-processing network in the macaque brain. Frontiers in Neuroscience 9:113. https://doi.org/10.3389/fnins.2015.00113
Wed Apr 28	Lecture 39	Hormones and behavior: the ring dove reproductive cycle (Lehrman) and seasonal song plasticity	Lehrman DS (1955). The physiological basis of parental feeding behavior in the ring dove (Streptopelia risoria). Behaviour 7:241–286. https://doi.org/10.1163/156853955X00094 Gillespie MJ, Haring VR, McColl KA, Monaghan P, Donald JA, Nicholas KR, Moore RJ, Crowley TM (2011). Histological and global gene expression analysis of the 'lactating' pigeon crop. BMC Genomics 12:452. https://doi.org/10.1186/1471-2164-12-452 Martinez-Vargas MC, Erickson CJ (1973). Some social and hormonal determinants of nest-building behaviour in the ring dove (Streptopelia risoria). Behaviour 45:12–36. https://doi.org/10.1163/156853974X00570 Cheng MF (1986). Female cooing promotes ovarian development in ring doves. Physiology & Behavior 37:371–374. https://doi.org/10.1016/0031-9384(86)90248-9
Fri Apr 30	Lecture 40	Frontiers: optogenetics, wireless recording, and causal tests of neural circuits	Luis A Bezares-Calderón; Jürgen Berger; Sanja Jasek; Csaba Verasztó; Sara Mendes; Martin Gühmann; Rodrigo Almeda; Réza Shahidi; Gáspár Jékely (2018). Neural circuitry of a polycystin-mediated hydrodynamic startle response for predator avoidance. eLife 7: e36262. https://doi.org/10.7554/elife.36262 Jessica Cande; Shigehiro Namiki; Jirui Qiu; Wyatt Korff; Gwyneth M Card; Joshua W Shaevitz; David L Stern; Gordon J Berman (2018). Optogenetic dissection of descending behavioral control in Drosophila. eLife 7: e34275. https://doi.org/10.7554/elife.34275 Hidehiko K Inagaki; Yonil Jung; Eric D Hoopfer; Allan M Wong; Neeli Mishra; John Y Lin; Roger Y Tsien; David J Anderson (2014). Optogenetic control of Drosophila using a red-shifted channelrhodopsin reveals experience-dependent influences on courtship. Nature Methods 11(3): 325-332. https://doi.org/10.1038/nmeth.2765 Georg Nagel; Tanjef Szellas; Wolfram Huhn; Suneel Kateriya; Nona Adeishvili; Peter Berthold; Doris Ollig; Peter Hegemann; Ernst Bamberg (2003). Channelrhodopsin-2, a directly light-gated cation-selective membrane channel. Proceedings of the National Academy of Sciences 100(24): 13940-13945. https://doi.org/10.1073/pnas.1936192100

Final Word

This course is hard, and it is designed to be. Neuroethology asks you to think like a biologist, an engineer, and a skeptic at the same time. If you do the reading, show up to lecture and lab, and start early, you will leave with a working understanding of how animals sense and act on their worlds. If you do not, the schedule will catch up with you by the first exam. Plan accordingly.